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POWER ELECTRONICS

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1. INTRODUCTION TO POWER ELECTRONICS

Electronic components and devices began with the invention of the semiconductor diode and bipolar transistor. Every device we use today has evolved over the years from these two basic semiconductor silicon devices. But as well as diodes and transistors, other types of semiconductor devices were invented to start the beginning of a new age of power electronics. These new devices belong to a family of **thyristors** used for the switching and control of electrical power and energy, either as an input or an output.

As its name implies, power electronics is a branch of electrical engineering that combines together the fields of solid-state electronics, electrical power, and control theory. Power electronics systems, can be sub-divided into two basic types of electronic elements. Power semiconductors, which do the actual switching of voltages and currents, and microelectronic control which monitors external conditions and tells the power semiconductors when to operate.

There are now several types of power semiconductor devices, such as *Silicon Controlled Rectifiers* (SCR), *Triacs* (Triode AC's), *Diacs* (Diode AC's) and *UJT's* (Unijunction Transistor). These electronic devices are all capable of operating as very fast solid state switches for the control of high voltages and currents into the kiloamperes and kilovolts range.

Today, power electronic systems and components are used everywhere with typical applications including: AC and DC motor drives, heating and lighting controls, phase control, electric vehicles, solid-state power supplies, battery chargers, controlled rectifiers, inverters, power switching devices, IOT, etc, etc, the list is endless with the silicon controlled rectifier being the most popular for such applications.

But it is also important to note that all the power semiconductor devices when used for power electronics and in power control applications are always operated as fast acting on/off switches to convert energy from one form to another, they are not amplifying devices.

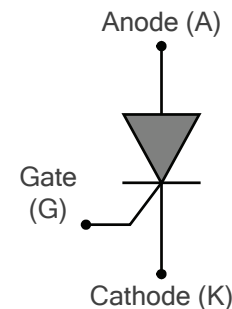
Power Electronics uses solid state semiconductor devices for high-power switching applications

2. THE SILICON CONTROLLED RECTIFIER

The **Silicon Controlled Rectifier**, (**SCR**) is the main switching device of the thyristor family, and as such is known more commonly as simply, a *Thyristor*. The SCR is a three-terminal solid state semiconductor device, hence the “*silicon*” part of its name. It requires a positive gate signal or pulse to turn it “ON”, the “*controlled*” part of the name and once “ON” it behaves like a rectifying diode, the “*rectifier*” part of its name.

While the junction diode is constructed as a two layer (p-n) rectifying device, and the bipolar transistor as a three layer (n-p-n or p-n-p) amplifying device. The silicon controlled rectifier is constructed as a four layer (p-n-p-n) ON/OFF switching device that contains three p-n junctions within its construction.

FIGURE 1. SILICON CONTROLLED RECTIFIER SYMBOL

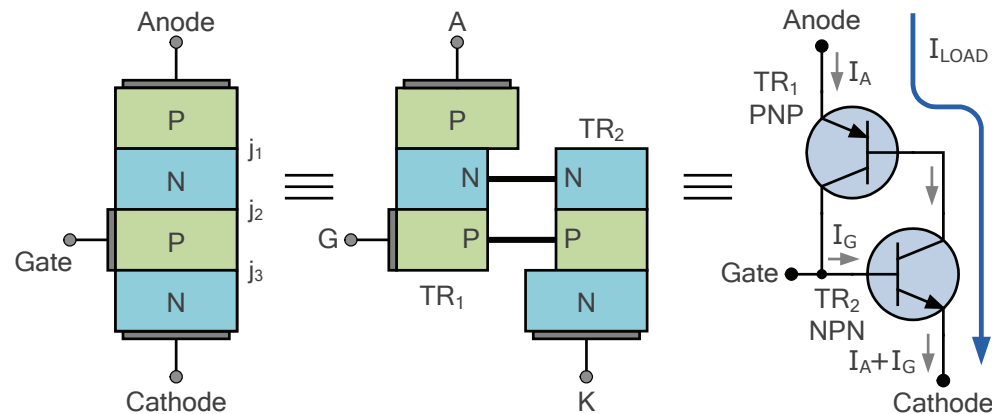


As shown in Figure 1. the silicon controlled rectifier (thyristor) has three connecting terminals labelled as: “Anode”, “Cathode” and “Gate”. It operates as a unidirectional device, similar to the diode, by passing current through itself from Anode to Cathode in one direction only. Hence its schematic symbol is similar to that of a conventional diode.

But unlike the diode, the thyristor can be made to operate as either an open-circuited switch when reverse biased, or as a rectifying diode when the thyristors gate is triggered. In other words, thyristors operate as on/off switching devices, but they cannot be used for amplification as a single gate pulse is only required to turn it fully-on into saturation.

Since the silicon controlled rectifier is a 4-layer device, it can be thought of as an n-p-n bipolar transistor with an extra p-layer added on to the end. The operation of the silicon controlled rectifier can best be explained by assuming that it is made up of a pair of tightly couples transistor layers connected back-to-back as a pair of complementary regenerative switches to produce a self-latching action. This simple back-to-back two-transistor analogy is shown in Figure 2.

FIGURE 2. SIMPLE TWO-TRANSISTOR MODEL OF AN SCR



The two transistor analogy of Figure 2. shows a PNP and an NPN transistor interconnected together to form a regenerative feedback pair. The n-type emitter of TR_2 becomes the device's Cathode (K), while the p-type emitter of TR_1 is the device Anode (A).

The collector current of the NPN transistor TR_2 feeds directly into the base of the PNP transistor TR_1 , while the collector current of TR_1 feeds into the base of TR_2 . That is, the collector current of transistor TR_1 becomes the base drive for TR_2 , and vice versa. The combined loop gain of this configuration will be equal to the product of the two individual transistors gains.

Clearly this two transistor analogy will not allow current to flow from the cathode to the anode terminal, but only when the anode is positive with respect to the cathode. Then these two theoretically inter-connected transistors rely heavily upon each other for conduction as each transistor gets its base-emitter current from the other's collector-emitter current. So until the NPN transistor TR_2 is given some base current to start conduction, nothing can happen even if an Anode-to-Cathode voltage is present.

When the anode terminal is negative with respect to the Cathode and the Gate terminal is open, the central n-p junction (j_2) becomes forward biased, but the two outer p-n junctions (j_1 and j_3) are reversed biased, thus the device behaves very much like an ordinary diode blocking reverse current flow.

If the Anode terminal is made more positive with respect to the Cathode and the Gate terminal is open, the two outer p-n junctions (j_1 and j_3) are now forward biased but the centre n-p junction (j_2) is reverse biased but any forward current is also blocked as the device is acting as an open-circuit switch.

However, if a positive current (I_G) is injected into the Gate terminal which is effectively the base of the NPN transistor TR_2 , the resulting collector current flows in the base of transistor TR_1 . This in turn causes a collector current to flow in the PNP transistor, TR_1 which increases the base current of TR_2 and so on.

Very rapidly the two transistors force each other into a permanent conducting state since they are connected as a regenerative feedback loop which cannot stop once started.

The SCR has four layers of alternating P-type and N-type semiconductor materials with silicon used as the intrinsic semiconductor

The electric current now flowing through the device between the Anode and the Cathode terminals is limited only by the impedance of the external circuit so long as current through the device remains above the holding current.

This regenerative switching action forcing the silicon controlled rectifier into conduction always occurs even if the gate voltage is applied permanently, or only momentarily as a gate pulse. Once triggered into conduction, and passing current in the forward direction (anode positive), the gate loses all control of the device due to the regenerative latching action of the two interconnected transistors.

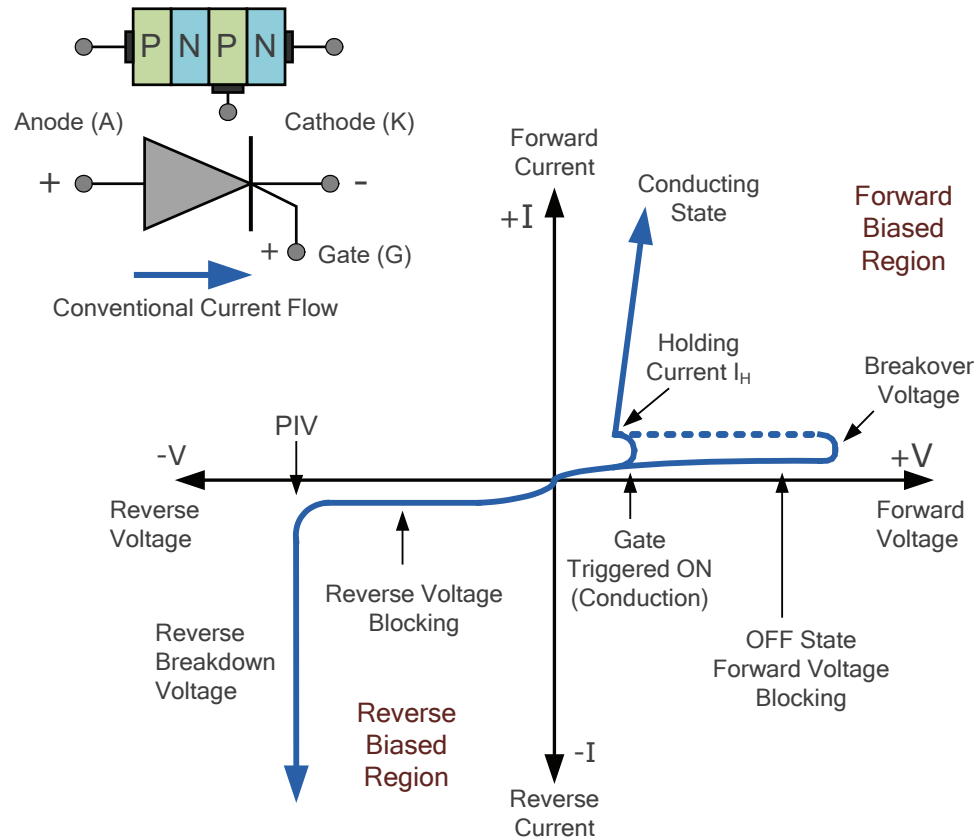
Note that the application of any additional gate signals or pulses after regeneration has been initiated will have no effect because the SCR is already conducting and fully-on.

The only way to turn a silicon controlled rectifier "OFF" (commutation) again is to reduce the anode-to-cathode voltage across it to zero (0) or its minimum holding current, (I_H) voltage value. This will cause both back-to-back transistors to turn off and remain in an off state until another Gate voltage is applied to the Gate terminal. The term 'turn-off' implies that a fully conducting device has returned back to its non-conducting forward blocking state.

Then we can see that a silicon controlled rectifier (SCR) has two possible states: a high impedance “off” state and a low-impedance “on” state. Under normal steady-state conditions an SCR can only be triggered on by the application of a positive drive signal, pulse, or supply of sufficient direction and amplitude to its “Gate” terminal.

The operating voltage-current I-V characteristics curves for the operation of a Silicon Controlled Rectifier are given in Figure 3.

FIGURE 3. I-V CHARACTERISTICS OF A SILICON CONTROLLED RECTIFIER



From the I-V characteristic curve of Figure 3, we can see that in the forward-biased (anode positive, cathode negative) region, the device remains in a non-conducting state, until either it is triggered “on” by a suitable gate pulse or the forward voltage reaches a high enough value to cause a forward breakover condition to occur. Once the breakover voltage is reached, the SCR conducts permanently.

In the reverse-biased (anode negative, cathode positive) region, the SCR blocks the flow of reverse current until at some high enough reverse voltage value, called the peak inverse voltage (PIV) rating, avalanche breakdown occurs and the SCR conducts without the application of any Gate signal.

This is an important negative characteristic of the silicon controlled rectifier, as SCR’s can be unintentionally triggered into conduction by a high reverse over-voltage as well as high temperature or a rapidly rising dv/dt voltage such as a voltage transient or spike.

One of the main advantages of using a silicon controlled rectifier as a switch is that it is a current operated device with a very high current gain, so a small Gate current pulse or signal can control a much larger Anode load current.

3. USING THE SCR WITH A DC SUPPLY

The operation of an SCR makes them suitable for use in medium to high-voltage AC power control applications, such as lamp dimming, regulators and motor control. But they can also be used in DC circuits as a crowbar device for overvoltage protection.

However, for SCR’s used in circuits fed from DC supplies, natural commutation condition cannot occur as the DC supply voltage is continuous and steady-state, so some other way to turn “off” the SCR must be provided.

There are two common methods for turning an SCR off at the appropriate time, current interruption and forced commutation. Both methods require reducing the anode-to-cathode voltage to zero, or by reducing the forward current to below the holding value and turning off the SCR because once triggered it will remain forever conducting.

FIGURE 4. GATE TRIGGERING BY A DC SIGNAL OR PULSE

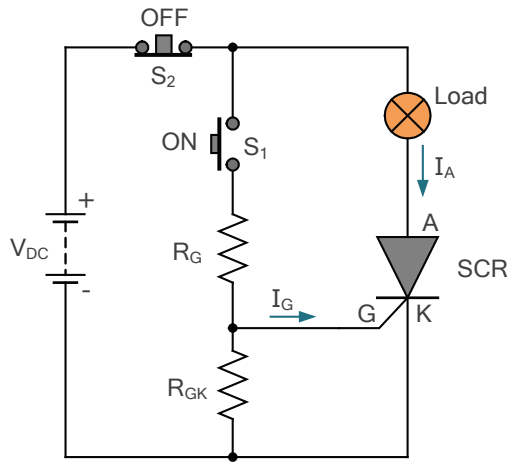


Figure 4. shows a simple “on-off” firing circuit which uses resistive triggering and the SCR as a switch to control a lamp load. It could also be used as an on-off control circuit for a motor, heater or some other such DC load.

The SCR is forward biased and triggered into conduction by briefly closing the normally-open “on” push button, S_1 . This briefly connects the SCR’s Gate terminal to the DC supply via the resistor, R_G .

Once the SCR has been triggered into conduction, it self-latches as before and stays conducting even when the push button is released due to the presence of the DC supply. The SCR will remain in its “on” state so long as load current through it remains above its holding current.

Additional operations of push button, S_1 will have no effect on the circuits state as once “latched” the Gate loses all control. The SCR is now turned fully “on” (conducting) allowing full load current to flow through the device in the forward direction and back to the battery supply.

As the SCR has self-latched into conduction, the circuit can only be reset (off) by interrupting the DC supply and reducing the SCR’s Anode current to zero or below its minimum holding current (I_H) value.

Operating the normally-closed “off” push button, S_2 breaks the circuit, reducing the circuit current flowing through the silicon controlled rectifier to zero, thus forcing it to turn “off” until the next application of another Gate signal. For example, the 2N650x series SCR, the typical holding current (I_H) value is about 18 mA.

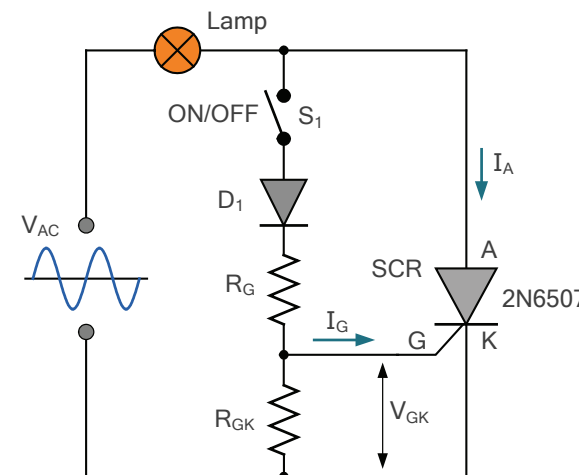
Note also that when an SCR is in its forward-conducting state, all three p-n junctions become forward biased and so it presents a minimum resistance to the forward Anode current flowing through it. For example, the 2N650x series I^2R voltage drop across the device when fully conducting is approximately 1.8 volts.

4. USING THE SCR WITH AN ALTERNATING AC SUPPLY

Since the silicon controlled rectifier has the ability to turn itself “off” whenever its forward current is reduced to zero or to below its minimum holding current value I_H , it therefore follows that when used on a sinusoidal AC supply the SCR will turn itself off at some value near to the crossover point of each and every half-cycle of the AC waveform, and as we now know, will remain in an off state until the application of the next positive Gate pulse.

Since an AC sinusoidal voltage supply is continually reversing in polarity from positive to negative and back again at every half-cycle. This periodic action allows the SCR to turn off automatically every 180° zero cross-over point of the positive half of the waveform. This effect is known as “natural commutation” and is a very important characteristic of the silicon controlled rectifier.

FIGURE 5. GATE TRIGGERING USING AN AC SIGNAL



The resistive SCR firing circuit of figure 5. is similar in design to the previous DC SCR circuit of Figure 4.

The difference here is that there is no additional “off” switch as it is generally not required with AC supplies.

The inclusion of diode D_1 helps prevent reverse bias being applied to the Gate terminal during the negative half-cycle from π to 2π .

During the positive half-cycle of the sinusoidal waveform of Figure 5, the SCR is forward biased but with switch S_1 open, zero gate current (I_G) is applied, so both the SCR and the lamp are off. On the negative half-cycle, the SCR is reverse biased as before so will remain off regardless of the switching condition of S_1 .

If switch S_1 of Figure 5. is now closed, at the beginning of each positive half-cycle the SCR is off. The positive going supply voltage increases upwards from zero at a rate determined by the supply frequency.

Diode D_1 becomes forward biased, and after a short time period there will be sufficient positive trigger voltage dropped across the resistive network of R_G and R_{GK} at the Gate terminal to turn the SCR and therefore the lamp "on".

The SCR is now latched fully-on for the remaining duration of the positive half-cycle but will automatically turn "off" again due to its natural commutation when the positive half-cycle ends and the Anode current falls below the SCR's holding current (I_H) value.

During the following negative half-cycle, the SCR remains fully-off and non-conducting until the next positive half-cycle is reached when the switching process repeats itself. The SCR is triggered into conduction once again as long as switch S_1 remains closed.

In the resistive triggering circuit of Figure 5. the fixed value resistor R_G effectively sets the firing angle of the SCR. Increasing R_G decreases the gate drive current and increases the firing angle.

Increasing the gate to cathode voltage above a certain threshold turns the SCR "on" and conducts current

Thus we can vary the angle of firing between 0° and 90° using this method by changing the resistive value of R_G , or by changing it to a suitable variable resistance. Resistor R_{GK} provides the Gate terminal with a negative bias during the negative half-cycle and keeps the SCR in a stable off state when there is no Gate drive.

So we can see then that the lamp (or any connected load) receives only half of the available power from the AC source as the SCR only conducts current during the positive half-cycle when it becomes forward biased.

Then the SCR acts like a rectifying diode during the negative half-cycle blocking current flow. The SCR will continue to supply half power to the lamp while switch S_1 is closed.

5. SCR PHASE CONTROL

We now know that a silicon controlled rectifier only conducts in one direction of a sinusoidal waveform under the control of the gate signal. This feature makes it ideally suited for use in phase control applications.

If it were possible to rapidly turn switch S_1 ON and OFF, so that the thyristor received its Gate signal at the "peak" (90°) point of each positive half-cycle, the device would only conduct for one half of the positive half-cycle. In other words, conduction would only take place during one-half of one-half of a sine wave. Thus this condition would result in the lamp receiving "one-fourth" or a quarter of the total power available from the AC source.

By accurately varying the timing relationship between the Gate pulse and the positive half of the sinusoidal waveform, the SCR can be made to turn on at any electrical angle, α (alpha) between 0° and 180° (0 to π) with respect to the positive half-cycle of the source voltage. This triggering angle α is commonly called the SCR's firing angle.

FIGURE 6. PHASE ANGLE TRIGGERING

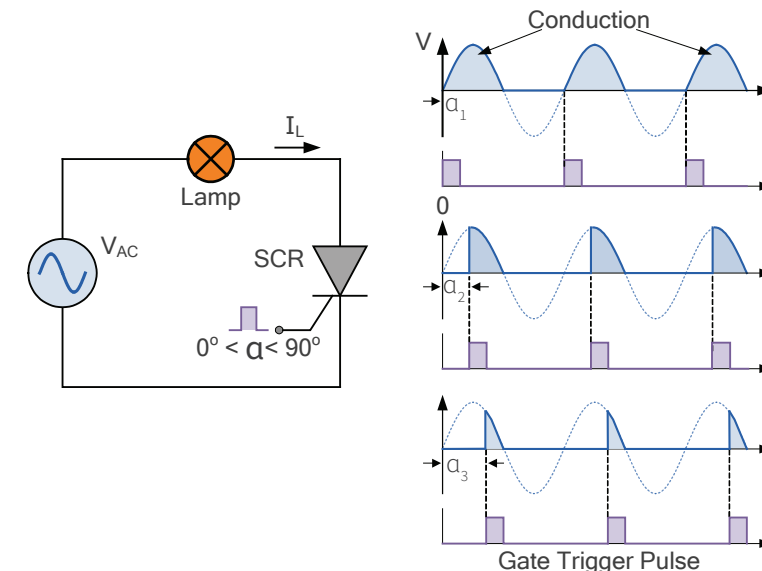


Figure 6. shows the effect of varying the firing angle of an SCR. At the start of each positive half-cycle the SCR is off due to natural commutation. If the SCR is triggered at the very beginning of the half-cycle ($\alpha_1 = 0^\circ$), the load (a lamp) will be “on” for the full duration of the positive half of the cycle.

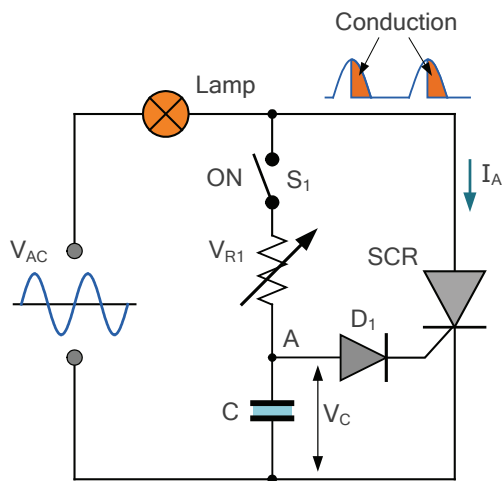
As the application of the gate trigger pulse increases along the half cycle ($\alpha = 0^\circ$ to 90°), the lamp will be illuminated for less time as the average voltage delivered to the lamp will be proportionally less, thus reducing its brightness.

Then phase control is achieved by varying the electrical angle at which the SCR triggered. That is by varying the firing angle α_f . For a half-wave circuit, the firing angle, α_f plus the conduction angle, α_c will be equal to 180 electrical degrees.

6. RESISTOR-CAPACITOR PHASE CONTROL

While the resistive triggering circuit of Figure 5. would work, the maximum firing angle is only 90° as the gate current is in-phase with the applied voltage. A larger variation to the firing angle can be obtained by changing the phase and the amplitude of the gate current using an RC triggering network.

FIGURE 7. RC PHASE ANGLE TRIGGERING CIRCUIT



The single-phase SCR circuit of Figure 7. uses a combination of a resistor and a capacitor to trigger the SCR into conduction.

During the positive half-cycle, the SCR is forward biased. Capacitor, C charges up via the variable resistor V_{R1} following the AC supply voltage waveform.

The Gate terminal triggers into conduction only when the voltage at point A has risen high enough to cause diode D_1 , to conduct.

At this point the capacitor discharges into the Gate of the SCR through the forward-biased diode turning it on. The time duration of the positive half of the cycle at which conduction starts is controlled by time constant of the passive RC combination.

Since the capacitor voltage V_C lags the supply voltage, V_{AC} by an angle α , this value is set by the variable resistor, V_{R1} . Increasing the value of V_{R1} has the effect of delaying the triggering voltage and current supplied to the Gate which in turn causes more lag in the conduction time increasing the RC time constant.

As a result, the fraction of the half-cycle over which the device conducts can be controlled between 0 and 180° , which means that the average power dissipated by the lamp can be adjusted. However, as we know, the SCR is a unidirectional device, only a maximum of 50% power can be supplied during each positive half-cycle, because it cannot conduct during the negative half-cycles when it is reverse biased.

There are a variety of ways to achieve 100% full-wave AC control using “thyristors”. One way is to include a single thyristor within a diode bridge rectifier circuit which converts AC to a unidirectional current through the thyristor while the more common method is to use two thyristors connected back-to-back in inverse parallel. A more practical approach is to use a single *Triac* as this device can be triggered in both directions, therefore making them suitable for AC switching applications.

7. THE TRIODE AC SWITCH

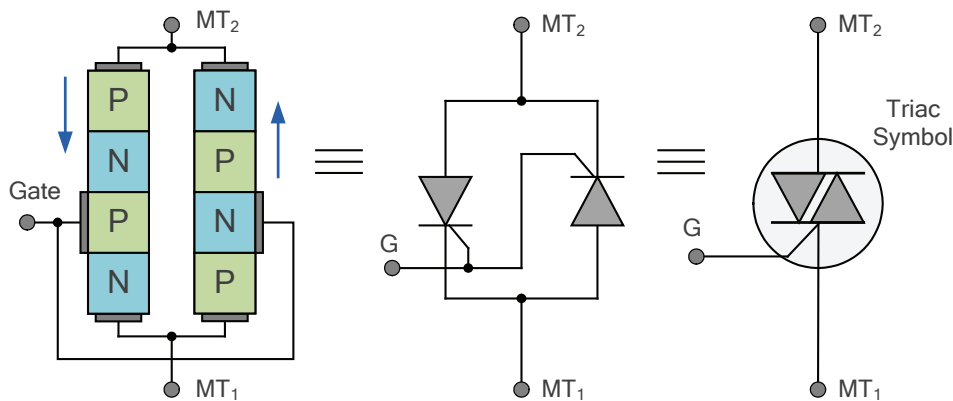
The “**Triode AC Switch**” or **Triac** for short, is another solid state semiconductor power switching device and which is also part of the thyristor family. The triac can be triggered into conduction using either a positive or a negative voltage applied to its anode terminal. Also, both positive and negative trigger pulses can be applied to its Gate terminal making it a bidirectional switching device.

Electrically, the triac behaves just like two silicon controlled rectifiers connected back-to-back in inverse parallel with respect to each other. Being a three-terminal device, these two back-to-back SCR’s share a common Gate terminal.

Since the triac conducts in both directions of a sinusoidal waveform, the concept of an Anode terminal and a Cathode terminal used to identify the main power terminals of an SCR are replaced with identifications of: MT_1 , for Main Terminal 1 and MT_2 for Main Terminal 2. The Gate terminal, G is referenced the same.

The triac's Gate terminal is associated with the MT_1 terminal, similar to the gate-cathode relationship of the silicon controlled rectifier. Thus current flow in either direction between MT_2 and MT_1 is initiated by applying a small current signal between MT_1 and the Gate. The construction and schematic symbol of a triac is shown in Figure 8.

FIGURE 8. TRIAC SYMBOL AND CONSTRUCTION

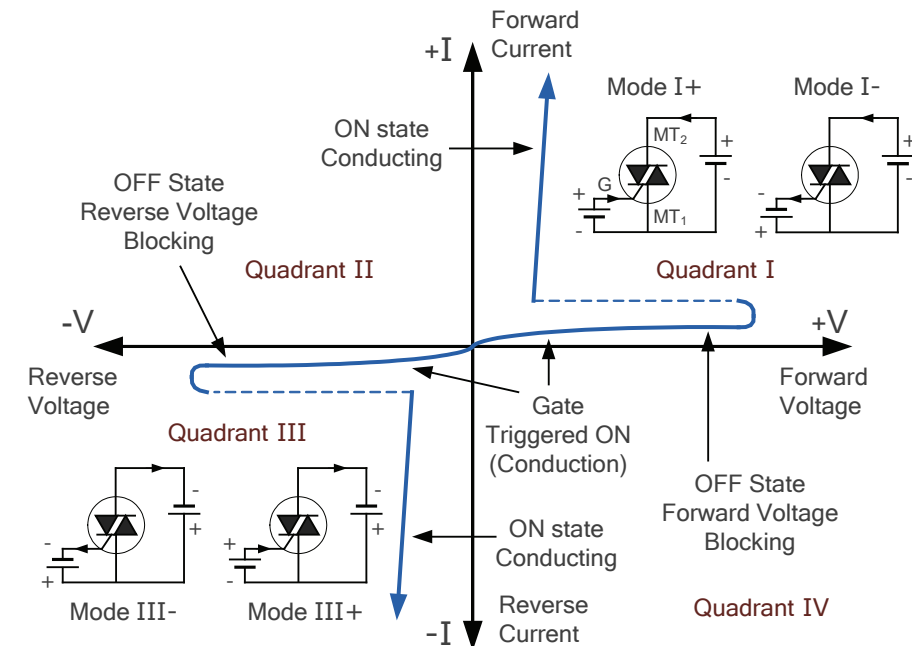


The triac is a 4-layer, p-n-p-n device in the positive direction and an n-p-n-p device in the negative direction. Unlike the silicon controlled rectifier, a triac can conduct current in either direction when triggered by a single gate pulse. Thus a triac has four possible triggering modes of operation as follows.

- I^+ Mode = MT_2 current positive (+ve), Gate current positive (+ve)
- I^- Mode = MT_2 current positive (+ve), Gate current negative (-ve)
- III^+ Mode = MT_2 current negative (-ve), Gate current positive (+ve)
- III^- Mode = MT_2 current negative (-ve), Gate current negative (-ve)

These four modes in which a bidirectional triac can be triggered into conduction in two quadrants is shown using the I-V characteristics curves of Figure 9.

FIGURE 9. TRIAC I-V CHARACTERISTICS CURVES



For standard AC phase control circuits such as lamp or motor control, the triac is usually triggered into conduction in Quadrant I by a positive gate current, labelled above as mode I^+ as the Gate and MT_2 polarities are always the same. This results in a more symmetrical switching action because the Gate is at its most sensitive. But it can also be triggered by a negative gate current, mode I^- .

Similarly, in Quadrant III triggering the Gate with a negative gate current, ($-I_G$) is also common. Resulting in mode III^- . Modes I^- and III^+ are, however, less sensitive configurations requiring a greater gate drive than the two common modes of I^+ and III^- . Hence, triac operation is commonly preferred in Quadrants I^+ and III^- only.

Also, just like SCR's, triac's also require a minimum holding current I_H to maintain conduction. Triac's still exhibit individual electrical characteristics such as different breakdown voltages, holding currents and trigger voltage levels exactly the same as we would expect from a single SCR device.

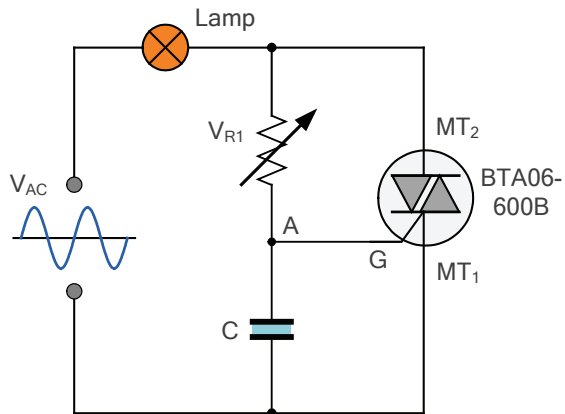
8. TRIAC PHASE CONTROL APPLICATIONS

As with the silicon controller rectifier, the triac can also be used as a simple static AC power switch providing a simple on/off zero-crossing switching function, or phase-control by varying its firing angle. In on/off power-control applications, a triac must change state from a conducting state to a blocking state twice per cycle, at each zero-crossing point.

Since the Gate trigger voltage is derived from the same AC power source, there is an inherent synchronisation of the Gate triggering pulse and the corresponding AC waveform. Thus the firing angle of the triac can be varied in both half-cycles using the previous RC phase-shifting network.

The basic phase triggering circuit of Figure 10. uses the triac in series with the lamp (or any load) across an AC sinusoidal supply. The variable resistor, V_{R1} controls the amount of phase shift on the Gate of the triac which in turn, controls the amount of voltage applied to the lamp load by turning it on at different times during the AC cycle.

FIGURE 10. TRIAC PHASE CONTROL



The triggering voltage value of the BTA06 triac is derived from the series RC combination of V_{R1} and C_1 . At the start of each half-cycle, C_1 charges up via the variable resistor, V_{R1} .

This continues until the voltage at point "A" is sufficiently high enough to trigger the triac into full conduction. This allows capacitor, C_1 to quickly discharge back into the gate of the triac ready for the next half cycle.

Note that once the triac is triggered into conduction and saturated, the triac effectively shorts out the gate triggering RC phase control circuit connected in parallel across it, and the triac takes control for the remainder of the half-cycle.

As we have seen above, the triac turns-off automatically at the zero crossing point of each half-cycle until the $V_{R1} - C_1$ triggering process starts again for the next half cycle.

However, because the triac requires differing amounts of Gate current in each switching mode of operation, for example **I*** and **III***, the switching action of the triac is therefore asymmetrical. This means that it may not trigger at the exact same phase angle or voltage point for each positive and negative half cycles as the non-polarised capacitor changes in different directions.

Also as with the silicon controlled rectifier, the triac's gate should be protected against transient over voltages and currents which may trigger it into conduction when off. Therefore the disadvantage of using this simple RC triggering network can be overcome by using a semiconductor triggering device such as the Diac.

9. THE DIODE AC SWITCH

The **Diode AC Switch**, or **Diac** for short, is another solid state bidirectional switching device. Unlike the SCR and the Triac, the diac has no Gate connection making it a two terminal device, labelled A_1 and A_2 . The diac is purely an electronic trigger diode which offers no control or amplification but produces a controlled breakdown switching action from either polarity of a suitable AC voltage supply.

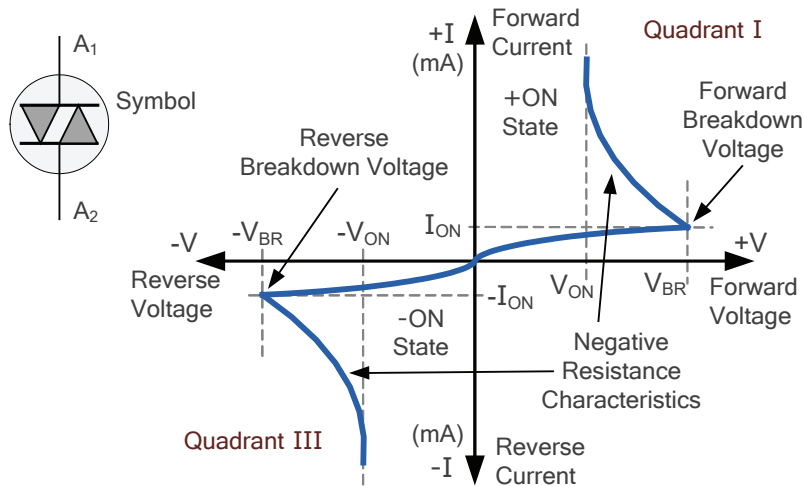
Internally the diac has a three-layer semiconductor structure similar in construction to the bipolar transistor. The difference being that the doping concentrations of the two pn-junctions are approximately the same resulting in a more symmetrical negative resistance characteristic in both directions when it begins conduction.

We can see from the diac I-V characteristics curves of Figure 11. that the diac blocks the flow of current in both directions until the applied voltage is greater than the device breakdown voltage V_{BR} .

At this point, instant voltage breakdown of the device occurs and the diac conducts heavily, in a similar way to a zener diode, passing a sudden pulse of voltage to the connected device.

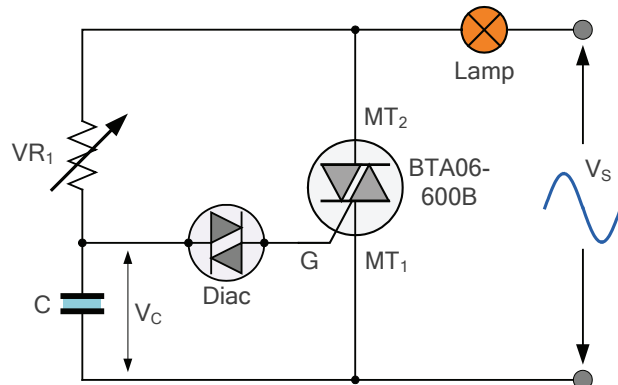
This V_{BR} point in either the first quadrant or the third quadrant is called the breakdown voltage or breakover voltage.

FIGURE 11. DIAC SWITCHING CHARACTERISTICS AND SYMBOL



Commercial diac's such as the DB3 or DB4 are designed with fixed breakdown voltage as low as about 30 to 40 volts and a high saturation voltage. Thus it can be used as a trigger device for SCR's and triac's in phase-triggering and variable power control applications.

FIGURE 12. AC POWER SWITCH WITH DIAC TRIGGERING



The advantage of using a diac as shown in Figure 12. is that they provide a sharper, more instant gate trigger pulse, as opposed to a steadily rising ramp switching voltage.

With reference to Figure 12. as the AC supply voltage increases at the beginning of each cycle, the capacitor charges up via the potentiometer, V_{R1} as before.

When the charge across capacitors plates has reached the breakdown voltage level of the diac, the diac conducts and the capacitor discharges into the gate through the diac. This sudden discharge produces a pulse of current, which fires the triac into conduction. The phase angle at which the triac is triggered can be varied using V_{R1} , which controls the charging rate of the capacitor.

Once the triac has been fired into conduction, it is maintained in its "ON" state by the load current flowing through it. The voltage across the resistor-capacitor combination is limited by the "ON" voltage of the triac.

The Diac is a gateless device designed to breakdown at a fixed voltage

This condition is maintained until the end of the present half-cycle of the AC supply. When the supply voltage enters its next half-cycle, the capacitor voltage again begins to rise (this time in the opposite direction) and the cycle of firing the triac repeats over again.

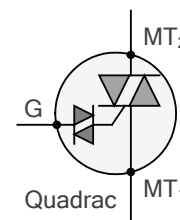
Then we have seen that when an increasing positive or negative half-cycle applied across the terminals of the diac reaches its trigger point, it undergoes avalanche breakdown exhibiting a negative-resistance characteristic which extends over a large range of current.

However, this means that whenever we want to use an SCR or triac for AC power control we may need a separate diac as well. Fortunately for us, a diac and a triac have already been combined together into a single switching device called a Quadrac.

10. THE QUADRAC

The **Quadrac** is basically a single semiconductor package with the *Diac* fabricated into the gate terminal of a *Triac* device. The advantage here is that it is one device and not two.

FIGURE 13. THE QUADRAC SYMBOL



The quadrac device is an internally triggered bidirectional AC switch and is gate controlled for either polarity of main terminal voltage which means it can be used in full-wave phase-control applications such as heater controls, lamp dimmers, and AC motor speed control, etc.

Like the triac, quadracs are a three-terminal semiconductor switching device labelled **MT2** for main terminal one (usually the anode), **MT1** for main terminal two (usually the cathode) and **G** for the gate terminal as shown in Figure 13.

The quadrac is available in a variety of package types depending upon their voltage and current switching requirements with the TO-220 package being the most common. The quadrac is designed to be an exact replacement for most triac devices.

11. THE UNIJUNCTION TRANSISTOR

The **Unijunction Transistor**, or **UJT** for short, is another solid state three-terminal device that can be used in gate pulse, timing circuits and trigger generator applications to switch and control either thyristors and triac's for AC power control type applications.

Although it uses the name of “*transistor*”, its switching characteristics are very different from those of a conventional bipolar junction or field effect transistor.

UJT's have unidirectional conductivity and negative impedance characteristics acting more like a variable voltage divider during breakdown so cannot be used for amplification but instead is as an on/off switching transistor.

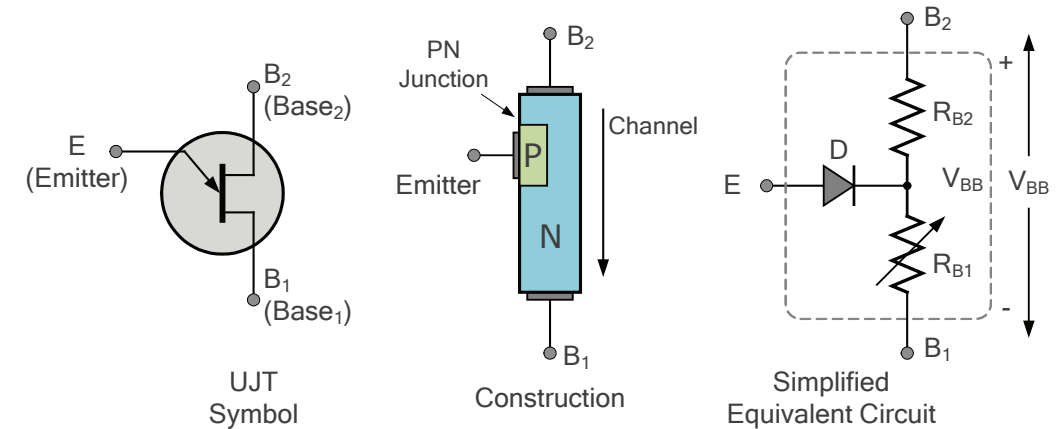
The Unijunction Transistor is a solid state triggering device with only one single pn-junction

Like n-channel FET's, the UJT consists of a single solid piece of n-type semiconductor material forming the main current carrying channel with its two outer connections marked as Base 2 (B_2) and Base 1 (B_1). Between Base 1 and Base 2 the unijunction transistor has the characteristics of an ordinary resistive device.

The third connection, confusingly named as the Emitter (E) is placed asymmetrically along the conducting channel and is formed by fusing the p-type material into the n-type silicon channel. The Emitter junction is positioned along the channel so that it is closer to terminal B_2 than B_1 .

An arrow is used in the UJT symbol which points towards the base indicating that the Emitter terminal is positive and the silicon bar is negative material. Figure 14. shows the symbol, construction, and equivalent circuit of the UJT.

FIGURE 14. THE UNIJUNCTION TRANSISTOR AND SYMBOL



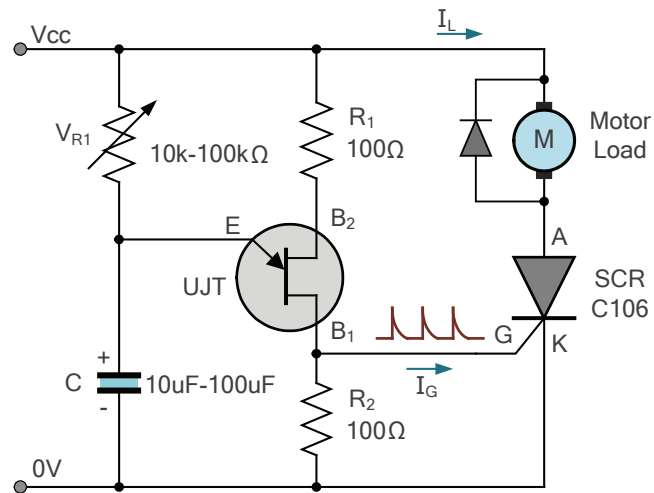
The symbol for the unijunction transistor is very similar to that of the junction field effect transistor or JFET, except that it has a bent arrow representing the Emitter (E) input. While similar in respect of their ohmic channels, JFET's and UJT's operate very differently and should not be confused.

Resistance R_{B1} is given between the Emitter, E and terminal B_1 , while resistance R_{B2} is given between the Emitter, E and terminal B_2 . As the physical position of the pn-junction is closer to terminal B_2 than B_1 the resistive value of R_{B2} will be less than R_{B1} . The static resistance of the channel is typically between about 4k Ω and 10k Ω 's for most common UJT's such as the 2N1671, 2N2646 or the 2N2647.

When used in a circuit, terminal B_1 is connected to ground and the Emitter serves as the input to the device. Suppose a voltage V_{BB} is applied across the UJT between B_2 and B_1 so that B_2 is biased positive relative to B_1 .

The most common application of a unijunction transistor is as a triggering device for SCR's and triacs but other UJT applications include sawtoothed generators, simple oscillators, phase control, and timing circuits. The simplest of all UJT circuits is the relaxation oscillator producing non-sinusoidal waveforms as shown in Figure 15.

FIGURE 15. UNIJUNCTION TRANSISTOR CIRCUIT



Initially, the capacitor is fully discharged but charges up exponentially through the variable resistor, V_{R1} . As the Emitter terminal of the UJT is connected to the capacitor, when the charging voltage V_C across the capacitor becomes greater than the forward voltage drop of the p-n junction, it behaves as a normal diode becoming forward biased triggering the UJT into conduction. The unijunction transistor is “on”.

At this point the Emitter to B_1 impedance collapses as the Emitter goes into a low impedance saturated state with the flow of Emitter current through resistor R_2 taking place. As the ohmic value of resistor R_2 is very low, the capacitor discharges rapidly through the UJT and a fast rising voltage pulse appears across R_2 .

When the voltage across the capacitor decreases below the holding point of the UJT's p-n junction (V_{OFF}), it turns off. No current flows into the Emitter junction so once again the capacitor charges up through V_{R1} . This charging and discharging process between V_{ON}

and V_{OFF} is constantly repeated producing a series of pulses to the SCR's Gate acting as an oscillator circuit.

The SCR of Figure 15. turns off due to the commutation action of the motor load. For AC operation using a triac the UJT triggering circuit can be fed from a suitable rectifier and smoothing circuit. This allows the UJT oscillator to be synchronised with the incoming mains supply.

The free-running oscillator circuit can be designed to produce 10's of trigger pulses per each half-cycle of the waveform. The variable resistance, V_{R1} controls the operating frequency of the UJT relaxation oscillator, which in turn controls the conduction angle of the triac. Clearly the triac will be fired into conduction on the first or second gate trigger pulse only, but this may be within a few degrees of the half-cycle giving full power to the connected load.

12. THE SOLID STATE RELAY

Solid State Relay's, or **SSR's**, are semiconductor equivalents of the electromechanical relay. Solid state relays are designed to switch and control both AC or DC currents without the use of moving parts by means of an SCR, TRIAC, or power switching transistor output, instead of the usual mechanical normally-open (NO) contacts.

Similar to an electro-mechanical relay, a small input voltage, typically 3 to 32 volts DC, can be used to control a much large output voltage, or current. For example, 240V, 10Amps. This makes them ideal for microcontroller, PIC and Arduino interfacing as a low-current, 3.3 or 5-volt signal from say a micro-controller or logic gate can be used to control a mains connected load.

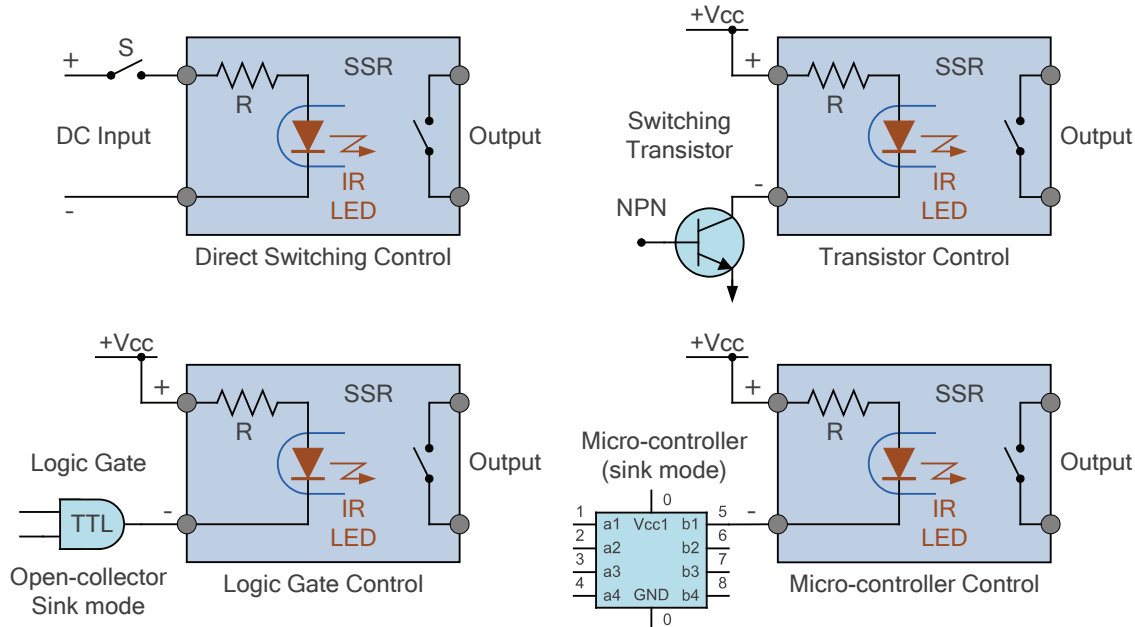
One of the main components of a solid state relay (SSR) is the Opto-isolator (also called an optocoupler) which contains one (or more) infra-red light emitting diode (IR LED) light source, and a photo sensitive device within a single package.

Solid State Relays use light to isolate their input and output with thyristors, triacs and transistors as switching elements

As the only connection between the input and output is a beam of light, high voltage isolation (usually several thousand volts) is easily achieved by means of this internal opto-isolation. Thus the output device of an opto-isolated solid state relay is turned “on” by energising this input LED with very low-voltage signal.

Figure 16 shows different ways to activate or turn “on” a typical solid state relay. A voltage greater than its minimum value (usually 3 volts DC) must be applied to its input terminals (equivalent to the electro-mechanical relay coil). This DC signal may be derived from a mechanical switch, digital logic gate or micro-controller port.

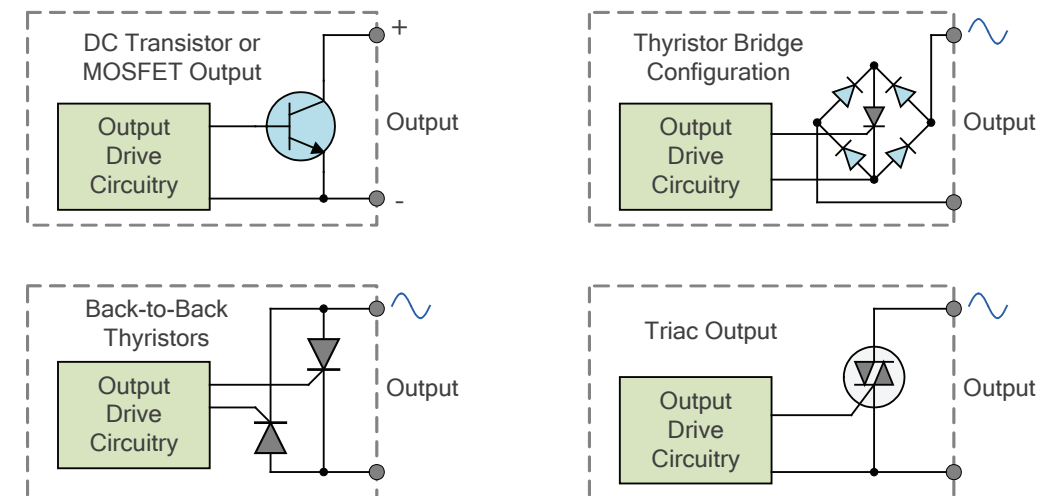
FIGURE 16. SOLID STATE RELAY DC INPUT CIRCUITS



The output switching capabilities of a solid state relay can be either AC or DC similar to its input voltage requirements. The output circuit of most standard solid state relays are configured to perform only one type of switching action giving the equivalent of a normally-open, single-pole, single-throw (SPST-NO) operation of an electro-mechanical relay.

For most DC SSR's the output switching device commonly used are power transistors, Darlington's, IGBT's or MOSFETs. For an AC SSR, the switching device is either a triac for low current applications or back-to-back SCRs for controlling very large amounts of power. As power switching devices, SCRs are preferred due to their high voltage and current capabilities.

FIGURE 17. SOLID STATE RELAY OUTPUT CIRCUIT TYPES

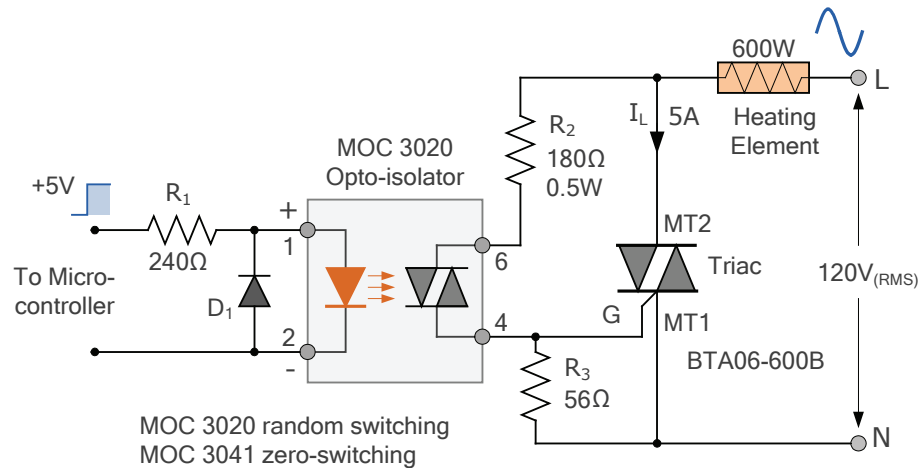


The most common application of solid state relays is in the switching of an AC load, whether that is to control the AC power for ON/OFF switching, light dimming, motor speed control or other such applications.

One of the biggest advantages of solid state relays over an electromechanical relay is its ability to switch “OFF” AC loads at the point of zero load current, thereby completely eliminating the arcing, electrical noise and contact bounce associated with conventional mechanical relays and inductive loads.

While there is a wide variety of commercially available solid state relays for switching high voltages and currents, we can also make our own inexpensive and simple version for switching lower AC loads such as a heater, lamp or solenoid using our previous knowledge about the triac as shown in Figure 18.

FIGURE 18. AC SWITCHING SSR



Here in Figure 18. the heating element is turned ON and OFF by turning the opto-isolator ON and OFF in response to a voltage output signal from a temperature controller or other such device. The load current of the heating element is switched using a triac.

As an opto-isolator only needs a small amount of input/control power to operate, the control signal could also be from a PIC, Arduino, Raspberry PI, or any other such micro-controller. The opto-isolator forms the basis of a very simple solid state relay application which can be used to control any AC mains powered load such as lamps and motors.

Here we have used the MOC 3020 which is a random switching opto-triac isolator. The MOC 3041 opto-triac isolator has the same characteristics but with built-in zero-crossing detection allowing the load to receive full power without the heavy inrush currents when switching inductive loads.

Diode D₁ prevents damage due to accidental reverse connection of the input voltage, while the 56-ohm resistor (R₃) shunts any di/dt currents when the triac is OFF eliminating false triggering. It also ties the gate terminal to MT1 ensuring the triac turns-off fully.

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